Assignment 1

- Why propagation model is required for wireless communication?
- Apply the different propagation models :Friis Propagation Loss Model, Log-Distance Propagation Loss Model (default), Log-Normal Shadowing Propagation Loss Model, International Telecommunication Union (ITU) Propagation Loss Model and Two-Ray Ground Propagation Loss Model.
- Plot the graph for signal strength perceived by sta1 for the above propagation model.
- Create network topology using mininet-wifi emulator with at least one or two access point, two or more station.

Why are Propagation Models Required?

In a virtual emulator like Mininet-WiFi, physical space does not actually exist; nodes are simply software interfaces assigned to virtual X,Y coordinates. A **Propagation Model** acts as the "physics engine" of the simulation.

It provides the mathematical formulas required to calculate how much a wireless signal degrades (attenuates) over a specific virtual distance due to factors like free space, obstacles, and ground reflection. Without a propagation model, a station 1,000 meters away would unrealistically receive the exact same perfect signal as a station 1 meter away, making realistic network simulation impossible.

The 5 Standard Propagation Models

- **Friis Propagation Loss Model:** Simulates an ideal, obstacle-free environment with a perfect line-of-sight.
- **Log-Distance Propagation Loss Model (Default):** A generalized model that uses a "path loss exponent" to simulate denser environments (like urban areas or offices) where the signal drops faster than in free space.
- **Log-Normal Shadowing Propagation Loss Model:** Builds on the Log-Distance model by adding a random Gaussian variable. This simulates random, unpredictable obstacles (like people moving or foliage).

- **ITU Propagation Loss Model:** An International Telecommunication Union standard specifically formulated for indoor environments and multi-floor building penetration.
- **Two-Ray Ground Propagation Loss Model:** Accounts for two paths: the direct line-of-sight signal and the signal reflecting off the ground. Highly accurate for longer distances.

Creating the Topology

Designing the Network Topology

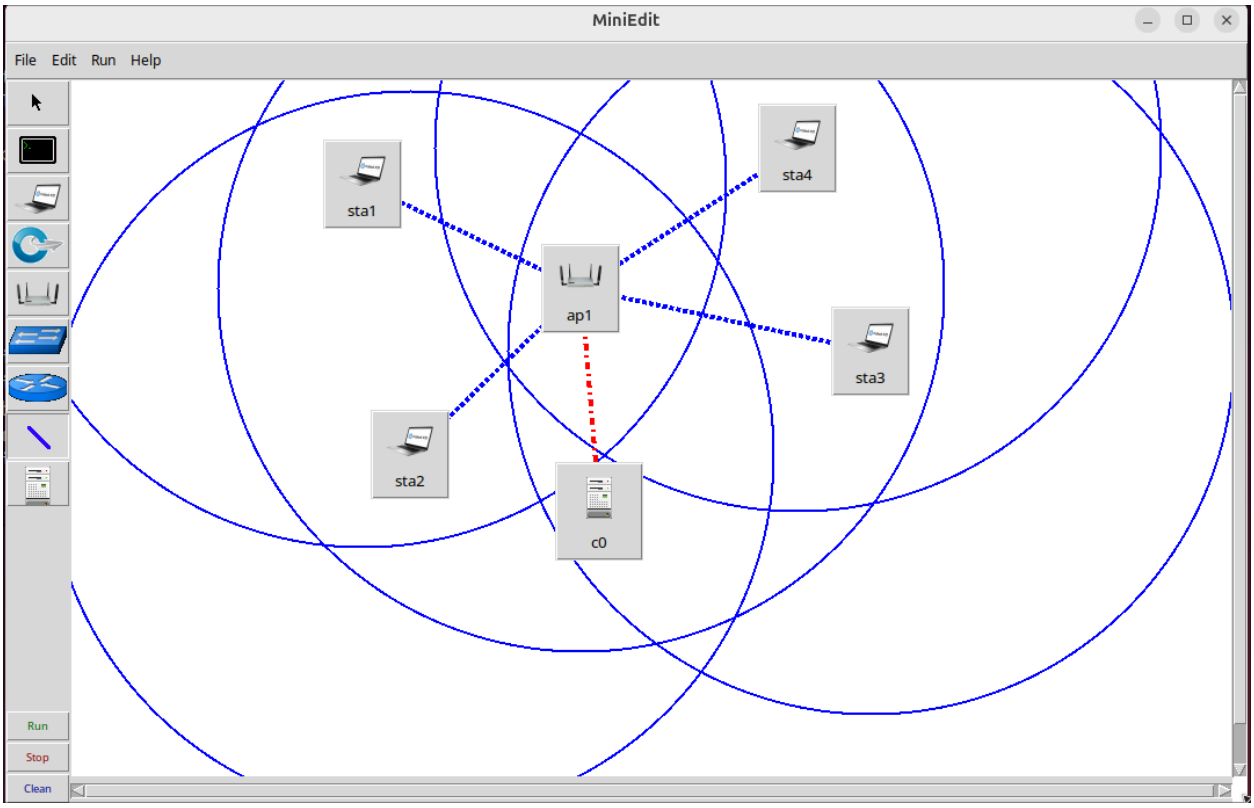
☐ **Step 1:** Launch the Mininet-WiFi GUI builder (MiniEdit) from your terminal:

```
sudo python3 mininet-wifi/examples/miniedit.py
```

☐ **Step 2:** From the left toolbar, drag and drop the following components onto the canvas:

- **1x Controller** (`c0`)
- **1x Access Point** (`ap1`)
- **4x Stations** (`sta1`, `sta2`, `sta3`, `sta4`) - Place them at varying distances from the AP.

☐ **Step 3:** Use the **NetLink** (Blue Line) tool to create a wired connection between the Controller (`c0`) and the Access Point (`ap1`).



Data Collection: Testing the 5 Official Models

We will now apply the 5 official models to your exported topology, run it, and record the signal strength.

- [ ] **Step 1:** Open your exported topology file in a text editor:

```
nano Documents/demo.py
```

- [ ] **Step 2:** Scroll down to the bottom. Look for the line that says `net.build()`.

Test 1: Friis Model

1. Right **above** `net.build()`, type the official Friis command from the documentation:

```
net.setPropagationModel(model="friis", sL=1)
```

2. Save the file and run it: `sudo python3 Documents/demo.py`

3. In the Mininet prompt, check the signal: `sta1 iw dev sta1-wlan0 link`
4. Record the **signal:** value (Recorded: `64 dBm`).
5. Type `exit` and clean up: `sudo mn -c`

Test 2: Log-Distance Model

1. Open `demo.py` again. Change the propagation line to match the official Log-Distance model:

```
net.setPropagationModel(model="logDistance", sL=1, exp=3)
```

2. Save, run the script, check the signal, and record the value (Recorded: `75 dBm`). Exit and cleanup: `sudo mn -c`.

Test 3: Log-Normal Shadowing Model

1. Open `demo.py`. Change the line to the official Log-Normal model:

```
net.setPropagationModel(model="logNormalShadowing", sL=1,
exp=3, variance=2)
```

2. Save, run, and record the signal value (Recorded: `78 dBm`). Exit and cleanup: `sudo mn -c`.

Test 4: ITU Model

1. Open `demo.py`. Change the line to the ITU model:

```
net.setPropagationModel(model="ITU")
```

2. Save, run, and record the signal value (Recorded: `72 dBm`). Exit and cleanup: `sudo mn -c`.

Test 5: Two-Ray Ground Model

1. Open `demo.py`. Change the line to the Two-Ray model:

```
net.setPropagationModel(model="twoRayGround")
```

2. Save, run, and record the signal value (Recorded: `68 dBm`). Exit and cleanup: `sudo mn -c`.

Plotting the Graph

